

COST-BENEFIT MEMO: MODEL SELECTION FOR PHASE 3

To: Dr De Freitas; Dr Marcus Reyes, Consultant Emergency Physician

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Re: Whether to adopt a more complex model (XGBoost) or keep the logistic-regression baseline

Setup: Same dataset and 80/20 stratified split as Week 6, random seed 42, outcome column excluded to prevent leakage

Verdict

Keep logistic regression as the model for Phase 3. The complex model (XGBoost) is not adopted. It wins overall accuracy by less than one point but performs worse on the metric that governs patient safety, and it gives up the per-patient explanation a clinician needs at the bedside.

How the candidates were chosen

Models were not tuned at random. Following good practice, the base versions of several models were compared first, and only then would effort go into optimising the best one. The shortlist was chosen from the shape of the data: about 55,000 patients, 226 features, a five-level ordinal target with severe class imbalance, and a hard requirement that each decision be explainable. For tabular data of this size and type, the sensible candidates are penalised logistic regression (handles wide, sparse features and is fully interpretable), gradient-boosted trees such as XGBoost (strong on mixed tabular data), and decision trees or random forests. Deep learning is not warranted here. The dataset is not large enough to need it, and it would remove the interpretability the clinical setting demands. This choice also matches the literature. The published triage models this project builds on, including e-triage and its later deployments, use gradient-boosted trees, which is exactly why XGBoost was selected as the complex model to test.

The benchmark

All three base models were compared on six measurable axes plus interpretability, on the same test set.

Six-axis benchmark - same split, seed 42 (Intel Core Ultra 9)

Model	Accuracy	Macro precision	Macro recall	Macro F1	ESI-1 recall	Train time (s)	Inference (ms/pred)	Interpretability
Logistic regression	0.685	0.666	0.476	0.518	0.25	31.6	0.003	High (per-patient)
Decision tree (d=6)	0.56	0.252	0.251	0.226	0.0	0.4	0.001	Medium (readable path)
XGBoost	0.693	0.692	0.441	0.47	0.062	29.0	0.034	Low (needs SHAP)

Times were measured on a Intel Core Ultra 9 processor and are indicative. The ratio between models is what matters, not the absolute figures.

The case for the complex model

Slightly higher overall accuracy. XGBoost reaches 0.693 against 0.685 for logistic regression, and higher macro precision (0.69 against 0.66), meaning its positive predictions are marginally more trustworthy.

Strong on this data type. Gradient-boosted trees are the standard high performer on tabular clinical data and captured non-linear interactions the linear model cannot.

Headroom with tuning. Only the base model was tested. Hyperparameter tuning or a recall-oriented objective could, in principle, lift its performance on the rare classes.

The case against

Worse where it matters. On the safety-critical numbers it is behind: macro F1 0.47 against 0.52, and recall on ESI 1, the sickest patients, of just 0.06 against 0.25. The complex model catches one in sixteen of the most urgent patients; the simple one catches four. For triage, that gap outweighs a fractional accuracy gain.

No bedside explanation. Logistic regression gives one reason per patient in a sentence: low oxygen and chest pain pushed this person up. XGBoost is 300 trees. A per-patient reason needs a separate tool (SHAP) and is still harder for a clinician to trust and act on in seconds.

Higher cost where it recurs. Training time was comparable on our hardware, but inference, the cost that repeats for every patient, every shift, was about eleven times slower per prediction (0.034 ms against 0.003 ms). For a live triage tool that recurring cost buys no safety benefit.

Risks of each choice

Keeping logistic regression risks leaving some achievable accuracy on the table, and the linear model may plateau as more features are added. Adopting XGBoost risks deploying a less safe and unexplainable model that a clinical team cannot audit, the exact pattern that caused the Epic Sepsis Model failure studied in Week 4. Between an interpretable model that is honest about its limits and a black box that is marginally more accurate overall but misses more critical patients, the interpretable model is the safer engineering choice for a first clinical pilot.

Recommendation and what would change it

Proceed to Phase 3 with logistic regression, using class weighting to raise recall on the urgent classes, and keep XGBoost as a documented comparison rather than the deployed model. This recommendation would change if a tuned gradient-boosted model could clearly beat logistic regression on ESI-1 recall while its explanations were made reliable for clinicians. Until then, the right model is the one the ED can understand and trust, not the one with the highest single score. With thirteen weeks and a first pilot, the goal is a safe, defensible baseline, not a fractional gain on a leaderboard.